

Beyond Scalar CISPO: Causal-Token Credit Allocation for Reasoning RL

Abstract

Long-chain reasoning reinforcement learning depends on stable policy-gradient updates over token sequences, but current clipped objectives usually broadcast the same sequence advantage and the same trust-region geometry to every generated token. This proposal studies a family of CISPO extensions that keep MiniMax’s clipped importance-sampling policy optimization (CISPO) as the stability backbone, while addressing what CISPO does not solve: token-level credit allocation under sparse sequence rewards. The first AD-CISPO prototype redistributed the upper clipping radius according to token saliency, but preliminary runs suggest that bound-only adaptation is a weak lever when few policy ratios reach the adaptive bound. The revised next-generation proposal is Credit-Weighted CISPO (CW-CISPO): a detached, mean-preserving token-weighted CISPO objective whose weights are computed from target-policy causal-tangent saliency, optionally gated by token entropy. Adaptive bounds remain as a guardrail and ablation rather than the sole use of saliency.

1 Problem and Motivation

Reinforcement learning has become a practical way to elicit mathematical and long-form reasoning from language models, especially when correctness can be checked by rule-based or programmatic rewards. PPO-style RLHF established clipped policy-gradient training as a scalable alignment method [1, 2], while GRPO removed the critic by using group-relative advantages and has been widely used for mathematical reasoning [3, 4]. More recent systems such as DAPO and MiniMax-M1 focus on stabilizing long-chain reasoning RL at scale [5, 6].

The unresolved problem is broader than scalar clipping: current outcome-reward RL largely broadcasts a sequence-level advantage to every generated token. In chain-of-thought-style reasoning, this is unlikely to be true. A token that introduces an intermediate quantity, selects an operation, or anchors a later conclusion may matter more than a formatting token or locally redundant phrase. CISPO fixes one important failure of PPO-style clipping by preserving dense gradients under clipped IS weights, but it still leaves token credit largely uniform. A single global clipping radius and a uniform token loss weight therefore create an undesirable tradeoff:

- if the radius is tight, important reasoning tokens may be prevented from moving enough when the reward signal identifies a better trajectory;
- if the radius is loose, low-value or unstable tokens receive the same update corridor, increasing variance and drift;
- if the same sequence advantage is broadcast to all tokens, deterministic filler tokens can absorb gradient mass that should be spent on uncertain or causally important reasoning decisions;
- if PPO/GRPO surrogate clipping suppresses token updates, the model can lose useful gradient signal exactly when the sampled policy has moved outside the trust corridor.

CISPO directly targets the third issue by clipping the importance-sampling (IS) weight itself and using the detached clipped weight to scale $\log \pi_\theta$, so clipped tokens still contribute gradients [6]. This proposal asks whether the remaining scalar-budget issue and the more general token-credit issue can be addressed without abandoning CISPO’s dense-gradient design.

2 Existing Solutions and Limitations

PPO and GRPO. PPO clips the policy-ratio surrogate to control policy updates while allowing multiple optimization epochs per batch [1]. GRPO adapts this family of methods to reasoning models by computing advantages from a group of sampled responses rather than a learned critic [3]. These methods are simple and effective, but their token-level clipping rules still use fixed trust-region hyperparameters.

DAPO and asymmetric clipping. DAPO improves long-chain reasoning RL with techniques such as a higher upper clip, dynamic sampling, token-level policy-gradient loss, and overlong reward shaping [5]. This is an important baseline because it shows that the upper clipping corridor matters for reasoning RL. However, DAPO still treats the upper corridor as a sequence-wide or globally configured scalar, not as a budget to be distributed across tokens.

CISPO. MiniMax-M1 proposes CISPO, which clips IS weights rather than clipping away token updates and reports improved RL efficiency in large-scale reasoning training [6]. CISPO is the strongest starting point for this proposal because it preserves per-token gradient density. Its limitation for this project is that it still broadcasts the same sequence advantage across tokens and uses a scalar IS clipping geometry.

Token-level credit assignment and reward redistribution. The broader RL literature has long treated delayed rewards as a credit assignment problem. RUDDER, for example, redistributes delayed returns through contribution analysis so that future rewards become easier to predict [17]. In LLM alignment, process supervision and process reward models explicitly replace pure outcome supervision with step-level feedback [27]. Recent token-level preference methods, including TDPO, TIS-DPO, SePO, T-REG, and TI-DPO, also argue that sequence-level preference optimization hides important token-level structure [18, 19, 20, 21, 22]. These works are important prior art for the revised proposal: they show that token-level weighting and token selection are active directions, so the novelty cannot simply be “use token weights.” The gap is instead online verifiable-reward RL built specifically on CISPO’s clipped-IS gradient, without external token reward labels, pairwise preference data, or a separately trained oracle selector.

Entropy and capacity views of RLVR credit. Recent RLVR analysis frames token credit as a reward-conditioned shift from the behavior policy toward a hindsight posterior, with token entropy upper-bounding where hindsight credit can reside [23]. This suggests that saliency alone is incomplete: a token can be locally objective-sensitive yet carry little learnable credit if it is already deterministic. A stronger CISPO extension should therefore test saliency-only weights against entropy-gated weights that focus update mass where the policy has both objective sensitivity and uncertainty capacity.

Attention-based saliency. Transformers route information through attention mechanisms [7], and attention patterns can provide useful structural signals. However, attention weights should not be treated as direct causal explanations of model behavior [8]. Attention sinks are a concrete example: initial or special tokens may receive high attention for non-semantic reasons [9]. AD-CISPO therefore uses frozen-reference attention only as an update-budget heuristic, not as a claim about human-interpretable explanation, and explicitly masks prompt and sink tokens.

Mechanistic attribution and gradient saliency. Mechanistic interpretability distinguishes observational signals such as attention patterns from intervention-based methods such as activation patching and causal tracing [10, 11]. Attribution patching approximates activation-patching effects with first-order gradients, making it more scalable but also susceptible to local-linearity failures [12, 13]. Classical attribution methods such as Integrated Gradients, DeepLIFT, and SmoothGrad further show that raw gradient saliency is scale-sensitive and noisy unless it is anchored to a baseline, averaged, or otherwise regularized [14, 15, 16]. These lessons motivate bounded, detached, mean-preserving weights and local smoothing: causal-tangent saliency should be treated as an optimization signal to ablate, not as a faithful explanation by default.

3 Research Gap

Existing reasoning-RL objectives answer two separate questions only partially:

1. How can policy-gradient updates remain stable under off-policy drift? PPO, GRPO, DAPO, and CISPO address this through clipped ratios or clipped IS weights.
2. Which generated tokens deserve a larger or smaller learning signal? Existing online RLVR methods largely still broadcast sequence-level rewards, even when they improve clipping, sampling, or normalization.

The gap is a CISPO-native token-credit objective for language-model RL: one that keeps CISPO’s detached clipped-IS gradient as the stability mechanism, but adds a bounded, falsifiable token-credit allocation layer derived from the current policy’s own computation on the sampled trajectory. This is different from process reward modeling, token-level DPO, or token selection in preference data because the signal is computed online from the RL rollout and used inside the CISPO gradient itself.

4 Proposed Solution

The revised proposal treats CISPO as a stable base objective and studies three increasingly strong ways to add token credit:

1. **AD-CISPO-Bound:** the original conservative variant, where saliency only redistributes the upper IS clipping radius.
2. **Credit-Weighted CISPO (CW-CISPO):** the lead next-generation variant, where saliency becomes a detached, bounded, mean-one multiplier on each token’s CISPO gradient while the scalar CISPO clip remains a guardrail.
3. **Hybrid CW-AD-CISPO:** an ablation where saliency controls both the token loss weight and the upper IS clipping radius, allowing the project to test whether credit allocation or trust-region allocation is the useful mechanism.

The lower IS bound remains inactive by default to match the MiniMax-style CISPO baseline. Each generated token receives a multiplier computed from a saliency extractor, and the multipliers are normalized so their mean over valid generated tokens is 1. This preserves the average per-sequence gradient scale or clipping budget while changing where the budget is spent. The first implementation used frozen-reference future-attention in-degree; the next-generation implementation should use target-policy causal-tangent saliency because attention in-degree can be dominated by sink tokens and because the target policy is the computation graph actually being optimized.

Core hypothesis. For reasoning tasks with sparse sequence-level rewards, CISPO’s dense-gradient property is necessary but not sufficient: learning efficiency is limited by uniform token-credit broadcast. Bounded, mean-preserving causal-token weights should improve reward efficiency relative to scalar CISPO if they move gradient mass toward objective-sensitive and uncertainty-bearing reasoning tokens. If shuffled weights, entropy-only weights, or uniform multipliers match the gain, then the method has not demonstrated a saliency-specific advantage.

What is novel. The proposal’s novelty is not the use of one particular saliency heuristic by itself, nor merely the fact of token weighting. The novel contribution is the combination of:

1. CISPO-compatible token-credit allocation that preserves the detached clipped-IS dense-gradient structure rather than replacing it with a preference-objective, process reward model, or PPO surrogate;
2. mean-preserving weight and multiplier normalization, so CW-CISPO reallocates gradient mass rather than increasing the total average per-sequence learning rate;
3. adaptive upper-clip redistribution as a controlled ablation and guardrail, not the only way saliency affects optimization;
4. an explicit saliency interface, where attention in-degree is a cheap baseline, causal-tangent saliency is the first objective-sensitive replacement, and entropy-gated causal saliency is the first capacity-aware variant;
5. sink-token diagnostics and guards, so punctuation or boilerplate attention sinks cannot silently absorb the adaptive learning budget;
6. falsifiable ablations that separate saliency from extra variance, entropy gating, asymmetric clipping, and reference-model overhead.

Novelty boundary after literature review. The proposal should not claim to invent token-level credit assignment, token selection, gradient attribution, or reward redistribution. Those ideas already exist in RUDDER, process supervision, token-level DPO variants, selective preference optimization, and mechanistic attribution. The narrower novelty claim is: *online, verifier-reward CISPO with detached clipped-IS weights, augmented by bounded mean-one token-credit weights derived from the target policy’s causal-tangent sensitivity and optionally gated by entropy*. If a prior method is found that already combines all of these elements, the fallback novelty becomes empirical: a controlled comparison showing whether CISPO-specific credit weighting improves reasoning RL where scalar CISPO is already strong.

5 Notation

Symbol	Meaning	Shape or domain
q	prompt/question	text
i	index of a sampled response within a prompt group	$i \in \{1, \dots, G\}$
o_i	response i in a sampled group	token sequence
t	token position in response i whose saliency, IS weight, or multiplier is being computed	$t \in M_i$
j	later token position in response i that attends back to position t	$j \in M_i, j > t$
u	summation index over valid generated positions in response i	$u \in M_i$
$o_{i,t}$	token at position t of response i	token id
G	number of responses sampled per prompt	positive integer
π_θ	current policy being optimized	language model
$\pi_{\theta_{\text{old}}}$	rollout policy that produced the batch	language model
π_{ref}	frozen reference model for optional KL and saliency baselines	language model
$r_{i,t}(\theta)$	IS weight $\frac{\pi_\theta(o_{i,t} q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} q, o_{i,<t})}$	positive scalar
$\hat{A}_i$	group-relative sequence advantage for response i	scalar
M_i	valid generated-token action positions for response i	token-position set
$S_{i,t}$	raw saliency for token $o_{i,t}$	nonnegative scalar
$H_{i,t}$	token entropy under the rollout or target policy	nonnegative scalar
$C_{i,t}$	optional capacity gate derived from entropy or uncertainty	nonnegative scalar
$m_{i,t}$	normalized token multiplier used for adaptive clipping	nonnegative scalar, mean 1 over M_i
$w_{i,t}$	detached normalized token-credit weight used in CW-CISPO	nonnegative scalar, mean 1 over M_i
$\epsilon_{\text{low}}^{IS}$	lower IS clipping radius	nonnegative scalar
$\epsilon_{\text{high}}^{IS}$	scalar upper IS clipping radius	positive scalar
$\epsilon_{i,t,\text{high}}^{IS}$	token-adaptive upper IS clipping radius	positive scalar

Index conventions. For each prompt, the model samples G responses; i selects one of them. Within response i , positions t , j , and u all refer to token indices in the same sequence, but play different roles: t is the position whose saliency or clipping budget is being assigned; j is a *later* position whose attention query looks back at t ; and u is a dummy index used only when averaging or summing over all valid generated positions in M_i .

6 Baseline: MiniMax-Style CISPO

The CISPO baseline starts from the off-policy REINFORCE objective with the IS weight treated as a stop-gradient coefficient:

$$J_{\text{ISPG}}(\theta) = \mathbb{E} \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \text{sg}(r_{i,t}(\theta)) \hat{A}_i \log \pi_\theta(o_{i,t} | q, o_{i,<t}) \right]. \quad (1)$$

CISPO clips the IS weight itself:

$$\hat{r}_{i,t}(\theta) = \text{clip} \left(r_{i,t}(\theta), 1 - \epsilon_{\text{low}}^{IS}, 1 + \epsilon_{\text{high}}^{IS} \right), \quad (2)$$

and optimizes

$$J_{\text{CISPO}}(\theta) = \mathbb{E} \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \text{sg}(\hat{r}_{i,t}(\theta)) \hat{A}_i \log \pi_{\theta}(o_{i,t} \mid q, o_{i,<t}) \right]. \quad (3)$$

A faithful baseline for this proposal sets the KL coefficient to zero and avoids an active lower IS-weight bound, following the MiniMax-M1 description of CISPO [6]. This isolates the effect of the upper clipping budget.

7 Bound-Only AD-CISPO Objective

The original AD-CISPO variant replaces the scalar upper clipping radius with a token-adaptive upper radius:

$$\epsilon_{i,t,\text{high}}^{IS} = \epsilon_{\text{high}}^{IS} m_{i,t}. \quad (4)$$

The clipped weight becomes

$$\hat{r}_{i,t}^{\text{AD}}(\theta) = \text{clip}(r_{i,t}(\theta), 1 - \epsilon_{\text{low}}^{IS}, 1 + \epsilon_{i,t,\text{high}}^{IS}), \quad (5)$$

and the objective is

$$J_{\text{AD-CISPO}}(\theta) = \mathbb{E} \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \text{sg}(\hat{r}_{i,t}^{\text{AD}}(\theta)) \hat{A}_i \log \pi_{\theta}(o_{i,t} \mid q, o_{i,<t}) \right]. \quad (6)$$

The adaptive multiplier is computed from token saliency. Let M_i denote the set of generated-token action positions in response i that are eligible for adaptive clipping. It excludes prompt positions, padding, masked-out actions, special tokens, and any explicit sink-token categories removed by the saliency guard. Here u ranges over all positions in M_i to form the mean saliency in the denominator, while t is the position whose multiplier is being computed:

$$\tilde{m}_{i,t} = \frac{S_{i,t}}{\frac{1}{|M_i|} \sum_{u \in M_i} S_{i,u}}. \quad (7)$$

If all valid saliency values are zero or non-finite, $\tilde{m}_{i,t}$ falls back to 1 for valid tokens. If lower or upper multiplier bounds are used, they must be followed by renormalization or projection so that

$$\frac{1}{|M_i|} \sum_{t \in M_i} m_{i,t} = 1. \quad (8)$$

A plain clamp is insufficient because it changes the total clipping budget.

8 Credit-Weighted CISPO Objective

The proposed next-generation objective uses saliency as a token-credit weight rather than only as an upper-bound modifier. Starting from the CISPO clipped IS weight $\hat{r}_{i,t}(\theta)$, define a detached nonnegative token-credit weight $w_{i,t}$ satisfying

$$\frac{1}{|M_i|} \sum_{t \in M_i} w_{i,t} = 1, \quad w_{\min} \leq w_{i,t} \leq w_{\max}. \quad (9)$$

CW-CISPO optimizes

$$J_{\text{CW-CISPO}}(\theta) = \mathbb{E} \left[\frac{1}{\sum_{i=1}^G |M_i|} \sum_{i=1}^G \sum_{t \in M_i} \text{sg}(w_{i,t}) \text{sg}(\hat{r}_{i,t}(\theta)) \hat{A}_i \log \pi_{\theta}(o_{i,t} \mid q, o_{i,<t}) \right]. \quad (10)$$

This objective leaves CISPO’s clipped-IS guardrail intact, but changes the allocation of gradient mass across tokens. Mean-one normalization is critical: without it, a run could improve merely because it increased the effective learning rate. Detaching $w_{i,t}$ is also critical: the saliency pass should decide how to allocate credit for the sampled trajectory, not create a second implicit objective that backpropagates through the saliency extractor during the policy update.

The simplest saliency-only weight is

$$\tilde{w}_{i,t} = \left(\frac{S_{i,t}}{\frac{1}{|M_i|} \sum_{u \in M_i} S_{i,u} + \epsilon} \right)^{\gamma}, \quad (11)$$

followed by bounded projection to mean 1. The exponent $\gamma \in [0, 1]$ controls how sharply the weight follows saliency. A conservative first run should use $\gamma = 1$ with $w_{\min} = 0.25$ and $w_{\max} = 2.0$, matching the existing AD-CISPO multiplier range.

Entropy-gated causal-token credit. Recent RLVR credit analysis suggests that entropy marks where hindsight credit can reside, while saliency marks where the current objective is locally sensitive [23]. A capacity-aware variant should therefore combine both:

$$\tilde{w}_{i,t}^{\text{cap}} = (S_{i,t} + \epsilon)^{\gamma} (C_{i,t} + \epsilon)^{\eta}, \quad C_{i,t} = \frac{H_{i,t}}{\frac{1}{|M_i|} \sum_{u \in M_i} H_{i,u} + \epsilon}. \quad (12)$$

Here $H_{i,t}$ can be the rollout-policy or target-policy entropy for the next token distribution at position t . The ablation set should include saliency only ($\eta = 0$), entropy only ($\gamma = 0$), and the product gate ($\gamma, \eta > 0$). This prevents the proposal from confusing objective sensitivity with learnable credit capacity.

Hybrid clipping and weighting. The hybrid variant uses one multiplier for credit weights and another for the upper clipping radius:

$$J_{\text{Hybrid}}(\theta) = J_{\text{CW-CISPO}}(\theta) \quad \text{with} \quad \epsilon_{i,t,\text{high}}^{IS} = \epsilon_{\text{high}}^{IS} m_{i,t}. \quad (13)$$

This is intentionally not the first hypothesis. Preliminary AD-CISPO runs show that upper-bound adaptation affects only the subset of tokens whose ratios approach the bound. CW-CISPO is a stronger use of saliency because every token gradient is reallocated while CISPO’s scalar clip still limits off-policy variance.

9 Saliency Estimation

The first AD-CISPO prototype used exact frozen-reference future-attention in-degree. It remains a useful baseline because it is cheap relative to target-policy causal-tangent saliency, but it should no longer be treated as the primary extractor. For a selected reference layer ℓ and head h , let $a_{i,j,t}^{(\ell,h)}$ denote the causal attention probability from the query at token position j to the key/value position

t . Position t is the saliency target; position j is each later valid generated token that attends to t . The saliency assigned to token $o_{i,t}$ is

$$S_{i,t} = \frac{1}{|L_{\text{top}}|} \sum_{\ell \in L_{\text{top}}} \frac{1}{H_\ell} \sum_{h=1}^{H_\ell} \sum_{\substack{j \in M_i \\ j > t}} a_{i,j,t}^{(\ell,h)}. \quad (14)$$

Thus a token is salient when later valid generated tokens attend to it under the frozen reference model. The strict condition $j > t$ prevents self-attention from inflating saliency. In this estimator, M_i is used both as the source query set and target key set, so prompt, padding, special-token, and explicitly guarded sink-token positions do not participate in the saliency budget.

The implementation computes this quantity blockwise from reference-model post-RoPE query and key states. This avoids materializing a full batch of $T \times T$ attention matrices while preserving exact softmax attention weights for the selected layers, including causal, padding, and sliding-window masks.

A cheaper ablation is frozen-reference key-vector norm:

$$S_{i,t}^{\text{KV}} = \frac{1}{|L_{\text{top}}|} \sum_{\ell \in L_{\text{top}}} \|k_{i,t}^{(\ell)}\|_2. \quad (15)$$

It should be reported as a proxy baseline, not as the primary saliency definition.

10 Mathematical Failure Mode: Attention Sinks versus Causal Importance

The future-attention in-degree estimator is best understood as an attention-centrality score:

$$S_{i,t} = \sum_{\ell,h} \sum_{\substack{j \in M_i \\ j > t}} a_{i,j,t}^{(\ell,h)}. \quad (16)$$

It answers the question “how much do later generated tokens look back at position t ?” A CISPO credit-allocation method, however, ultimately needs a different quantity: how much changing the representation at position t would change the policy-gradient objective. These two quantities can diverge because attention mass is not a causal explanation of model behavior [8].

In a transformer attention layer, the value mixed into query position j is

$$z_{i,j} = \sum_{t \leq j} a_{i,j,t} V_{i,t}. \quad (17)$$

A large coefficient $a_{i,j,t}$ says that position j reads from position t , but it does not imply that $V_{i,t}$ changes the final answer, reward, or policy-gradient update. This distinction matters for attention sinks [9]: early separator or formatting tokens can attract many small attention weights from later positions for non-semantic reasons.

For example, suppose a response begins with a common reasoning preamble:

Okay, let’s try to figure out ...

Let t_c be the comma position. If many future layer-head-query terms assign a small amount of attention to this comma, the accumulated in-degree can dominate:

$$a_{i,j,t_c}^{(\ell,h)} \approx 0.02, \quad 400 \text{ contributing terms} \quad \Rightarrow \quad S_{i,t_c} \approx 8.0. \quad (18)$$

A genuinely reasoning-critical token may receive stronger attention from fewer positions:

$$20 \text{ contributing terms} \times 0.10 \Rightarrow S_{i,t_{\text{math}}} \approx 2.0. \quad (19)$$

Under attention centrality, the comma receives the larger score even though it may not be useful for solving the problem.

This creates a direct optimizer-level failure. AD-CISPO converts saliency into an upper clipping budget:

$$\epsilon_{i,t,\text{high}}^{IS} = \epsilon_{\text{high}}^{IS} m_{i,t}. \quad (20)$$

If the comma sink receives $m_{i,t_c} = 2$ and $\epsilon_{\text{high}}^{IS} = 0.2$, its upper IS bound becomes

$$1 + \epsilon_{i,t_c,\text{high}}^{IS} = 1 + 0.2 \cdot 2 = 1.4. \quad (21)$$

Thus the largest adaptive trust region can be allocated to punctuation in a boilerplate prefix rather than to the tokens whose representations influence the rewarded reasoning trajectory.

Causal-tangent saliency. A better first-generation replacement is an attribution-patching-style first-order estimator, which approximates causal intervention effects with a single backward pass [10, 11, 12]. Define a teacher-forced scalar objective over sampled completions:

$$F(\theta) = \sum_i |\hat{A}_i| \frac{\sum_{t \in M_i} \log \pi_\theta(o_{i,t} \mid q_i, o_{i,<t})}{|M_i|}. \quad (22)$$

Let $h_{i,t}^{(\ell)}$ be the residual-stream activation at selected layer ℓ and generated position t . A first-order Taylor expansion gives

$$F(h_{i,t}^{(\ell)} + \Delta) - F(h_{i,t}^{(\ell)}) \approx \nabla_{h_{i,t}^{(\ell)}} F \cdot \Delta. \quad (23)$$

The word “causal” is used in the intervention sense, not as a claim that the score is a complete causal proof. The ideal causal question is: if one could intervene on the hidden state at token position t while holding the rest of the computation fixed as much as possible, how much would the scalar objective F change? Exact activation patching answers this by replacing or perturbing $h_{i,t}^{(\ell)}$ and rerunning the downstream computation, but doing this for every token, layer, and rollout is too expensive inside RL training. The first-order term above is the tangent-space approximation to that intervention: it estimates the local effect of an infinitesimal do-intervention $\text{do}(h_{i,t}^{(\ell)} \leftarrow h_{i,t}^{(\ell)} + \Delta)$ on F . Thus the method is “causal” because it estimates the effect of changing an internal state on the objective, and “tangent” because it uses the local linearization rather than an explicit finite intervention.

This suggests a token saliency score

$$S_{i,t}^{\text{CT}} = \sum_{\ell \in L_{\text{top}}} \frac{\left| \nabla_{h_{i,t}^{(\ell)}} F \cdot \left(h_{i,t}^{(\ell)} - \frac{1}{|M_i|} \sum_{u \in M_i} h_{i,u}^{(\ell)} \right) \right|}{\sqrt{d_{\text{model}}}}. \quad (24)$$

Because tokenization can split one reasoning step across several adjacent subtokens, the training signal should use a locally smoothed score rather than the raw point estimate alone. The default smoothed score is

$$\bar{S}_{i,t}^{\text{CT}} = \lambda_0 S_{i,t}^{\text{CT}} + \lambda_1 S_{i,t-1}^{\text{CT}} + \lambda_1 S_{i,t+1}^{\text{CT}}, \quad \lambda_0 + 2\lambda_1 = 1, \quad (25)$$

with neighboring terms dropped and the weights renormalized at sequence boundaries or masked positions. This is not meant to blur across whole sentences; it is a tokenization-scale regularizer that reduces brittle single-token spikes, analogous in spirit to smoothing noisy gradient attributions [16].

The geometric intuition behind this score is that $\nabla_{h_{i,t}^{(\ell)}} F$ is the normal vector to the local tangent plane of the training objective with respect to the token-position representation. The centered activation

$$\bar{h}_{i,t}^{(\ell)} = h_{i,t}^{(\ell)} - \frac{1}{|M_i|} \sum_{u \in M_i} h_{i,u}^{(\ell)} \quad (26)$$

is a direction from the response’s average generated-token state toward the specific token’s representation. Their inner product measures how much the token-specific representation points in a direction that locally increases or decreases the scalar objective. Taking the absolute value treats both positive and negative local influence as importance: a token can matter because it supports the rewarded trajectory, or because the objective is strongly trying to move away from its current representation.

This differs from attention in-degree in three important ways. First, a token must be connected to the objective gradient; receiving attention is not enough. If $\nabla_{h_{i,t}^{(\ell)}} F \approx 0$, then $S_{i,t}^{\text{CT}}$ is small even when the token is an attention sink. Second, centering removes common sequence-level activation structure, so generic boilerplate magnitude is less likely to dominate. Third, summing over the final selected layers asks whether the token representation remains useful near the model’s decision surface, rather than whether it was merely read by later tokens in an earlier attention computation.

For CW-CISPO and AD-CISPO-Bound, the practical interpretation is:

allocate more CISPO learning pressure, and optionally more upper IS clipping budget, to tokens whose hidden states lie in directions to which the actual policy-gradient objective is locally sensitive.

This is why causal-tangent saliency is a better match to CISPO extension than attention centrality. The multiplier is no longer driven by how often a token is looked at; it is driven by a first-order estimate of how much the training objective would change if that token-position representation were perturbed.

The proposed next-generation extractor should therefore treat attention in-degree as a diagnostic baseline and use causal-tangent saliency, plus explicit sink-token guards, as the first serious replacement when attention sinks dominate the top-saliency table.

Target policy versus frozen policy. Causal-tangent saliency should use activations and gradients from the current target policy π_θ , not from the frozen reference policy π_{ref} . The reason is built into the definition of F : it is the teacher-forced objective whose log-probability terms are produced by the model being updated. The gradient

$$\nabla_{h_{i,t}^{(\ell)}} F \quad (27)$$

is therefore a sensitivity of the target policy’s computation graph. If the same score were computed on π_{ref} , it would estimate the effect of perturbing hidden states in a model that will not receive the CISPO update. That would reintroduce the same mismatch that made frozen-reference attention-saliency fragile: the saliency source could emphasize obsolete features or sink behavior from a policy that is no longer the object being optimized.

The frozen policy remains useful as a baseline and diagnostic. It can provide KL/reference metrics, future-attention in-degree ablations, and drift measurements. But the primary causal-tangent extractor should run after rollouts are collected and before the policy update, using the current training model π_θ on the sampled prompt-response sequences.

11 Implementation Plan

The minimum implementation should use the existing Curious training path and GSM8K reward setup [24]. The intended experimental model is a small reasoning-capable open model, such as the repository’s Qwen3-1.7B configuration, so that full ablations are affordable on a single H100.

Next-generation implementation flow: $\pi_{\theta_{\text{old}}}$ samples grouped responses $\rightarrow$ verifier computes rewards and group advantages $\rightarrow$ target policy is run teacher-forced on sampled responses $\rightarrow$ final-layer hidden states are captured $\rightarrow$ the advantage-weighted log-prob objective F is backpropagated to those hidden states $\rightarrow$ causal-tangent saliency $S_{i,t}^{\text{CT}}$ is computed and sink-guarded $\rightarrow$ saliency and optional entropy are normalized into $w_{i,t}$ $\rightarrow$ optional bound-only ablations also derive $m_{i,t}$ $\rightarrow$ token thresholds $\epsilon_{i,t,\text{high}}^{\text{IS}} = \epsilon_{\text{high}}^{\text{IS}} m_{i,t}$ $\rightarrow$ CW-CISPO loss uses detached token weights and detached clipped IS weights $\rightarrow$ metrics log reward, exact match, clipping, saliency shape, credit-weight shape, sink-guard behavior, and runtime overhead.

Implementation contract. A mathematically aligned CISPO-extension implementation should satisfy the following:

1. CISPO loss must use a detached clipped IS weight multiplying $\log \pi_{\theta}$, not a PPO/GRPO surrogate minimum.
2. Adaptive CISPO variants must be enabled only with the CISPO loss, unless explicitly naming and evaluating an adaptive-GRPO ablation.
3. MiniMax-style CISPO should set KL weight to zero and avoid an active lower IS-weight bound.
4. Token-adaptive clipping should change only the upper IS clipping radius unless a lower-bound ablation is deliberately introduced.
5. Token-credit weighting should use detached nonnegative weights $w_{i,t}$ projected to mean 1 per response; raw saliency values should not be multiplied directly into the loss.
6. CW-CISPO should first keep scalar CISPO clipping fixed. Bound redistribution should be introduced as a separate AD-CISPO-Bound or hybrid ablation.
7. Saliency tensors should align with the generated token whose IS weight is being clipped. In next-token prediction implementations, generated-token saliency must align with next-token log-probability terms, not the preceding hidden-state position.
8. Causal-tangent saliency should be computed from the target policy on the sampled completions, with the scalar objective using the same generated-token log-probability terms and group advantages that drive the CISPO update. Frozen-policy activations should be reserved for explicit ablations, not for the primary extractor.
9. The causal-tangent backward pass should differentiate only the saliency objective used to score token positions; the final CISPO update should still use the normal training loss and should not accidentally reuse stale gradients from saliency extraction.
10. Sink guards should be explicit and logged. Separator punctuation, whitespace, special tokens, and prompt tokens should either be removed from weight and multiplier assignment or capped at 1, while numeric tokens and mathematical operators should remain eligible unless ablated.
11. Claims about compute and memory must distinguish active-policy saliency overhead from frozen-reference overhead.

Causal-tangent extractor steps. The recommended first implementation is:

1. Add a saliency method flag such as `ad_cispo_saliency_method=causal_tangent`.
2. During trajectory processing, run the target policy teacher-forced over the sampled prompt-response sequences, using the same logits-minibatch boundary used for log-probability computation. This should use the current training model before the update step, not the frozen reference model.
3. Register hooks on the last four decoder layers and retain gradients for their hidden-state tensors at generated-token positions.
4. Form F from generated-token log-probabilities weighted by $|\hat{A}_i|$, with per-response length normalization so long completions do not dominate saliency solely by length.
5. Backpropagate F , compute $S_{i,t}^{\text{CT}}$ from gradient-centered-activation inner products, then detach and clear temporary gradients before the policy update.
6. Apply local smoothing, sink guards, and log the fraction of tokens and saliency mass removed by those guards.
7. Add mean-preserving token-credit-weight projection. Reuse the existing multiplier projection for bound-only and hybrid ablations, but keep $w_{i,t}$ and $m_{i,t}$ separately logged.

12 Alternative CISPO Extensions

CW-CISPO is the lead proposal, but the literature review suggests several other CISPO-native directions that solve problems CISPO has not yet addressed:

1. Entropy-Capacity CISPO. CISPO clips IS weights but does not decide which tokens can carry useful hindsight credit. Entropy-Capacity CISPO would keep the scalar CISPO objective and weight tokens by a bounded function of token entropy. This is cheaper than causal-tangent saliency and directly tests the hypothesis that reasoning gains come from uncertain decision points. Its weakness is that entropy measures capacity, not whether the token is actually relevant to the rewarded trajectory.

2. Causal-Token CW-CISPO. This is the main proposal: combine causal-tangent saliency with optional entropy capacity. It solves a more specific problem than entropy-only weighting: not only which tokens *can* change, but which token-position representations the current policy-gradient objective is locally sensitive to. Its cost is the extra teacher-forced backward pass.

3. Selective-CISPO. Instead of weighting all tokens, optimize only the top- p percent of tokens by saliency, entropy, or their product, while still using CISPO’s clipped IS coefficient. This is analogous in spirit to selective preference optimization [20], but applied online to RLVR rollouts rather than offline preference pairs. It could reduce memory and compute, but is riskier because hard selection can drop useful connective tokens.

4. Variance-Adaptive CISPO. CISPO clips IS weights to control variance, but the clipping radius is fixed. A variance-adaptive extension would tune $\epsilon_{\text{high}}^{IS}$ from observed ratio dispersion, clipped-token fraction, and advantage variance, rather than from saliency. This is less novel conceptually, but it may be a strong practical baseline because recent AD-CISPO runs show that adaptive saliency bounds rarely bind.

5. Step-Chunk CISPO. Instead of assigning weights at the subtoken level, group neighboring tokens into reasoning chunks using newline, equation, or sentence boundaries and apply one CISPO credit weight per chunk. This addresses token-sink artifacts and tokenization noise. It is less mechanistically fine-grained, but may better match process-supervision intuitions when the reward depends on reasoning steps rather than isolated subtokens.

The first experimental priority should be Causal-Token CW-CISPO versus Entropy-Capacity CISPO and scalar CISPO. If causal saliency does not beat entropy-only weighting, then the causal-tangent extractor is not yet adding useful credit information beyond uncertainty capacity.

13 Evaluation Plan

Datasets and tasks. The evaluation should use a staged math-reasoning suite rather than a single benchmark:

1. **GSM8K** should remain the first target, because it has a clear exact-answer reward and is already supported in Curious [24]. It is the right dataset for fast debugging, seed sweeps, and early reward-curve comparisons.
2. **SVAMP** should be used as an evaluation-only robustness check for arithmetic word-problem variations [25]. This helps test whether adaptive CISPO improves reasoning robustness rather than merely fitting GSM8K surface patterns.
3. **MATH and MATH-500** should be the next difficulty tier. MATH provides competition-style problems with step-by-step solutions [26], while MATH-500 is a smaller representative subset commonly used for cheaper repeated evaluation [27]. These tasks test whether the token-adaptive clipping signal transfers to longer derivations and more diverse mathematical domains.
4. **AIME/AMC-style held-out sets** should be reserved for final stress testing once answer extraction is stable. If broader olympiad-level coverage is needed, add the mathematics split of OlympiadBench as an optional evaluation-only benchmark [28].

Training should begin with GSM8K and expand only after scalar CISPO and adaptive CISPO have matched data loading, answer extraction, rollout count, and compute budgets. Harder datasets should be used primarily to measure transfer, not to tune the saliency or token-credit heuristic.

Baselines. The minimum comparison set is:

1. GRPO/PPO-style surrogate baseline.
2. DAPO-style asymmetric clipping or clip-high baseline.
3. MiniMax-style CISPO: no KL, no active lower IS bound, scalar upper IS radius.
4. AD-CISPO-Bound with causal-tangent saliency and sink guards.

5. CW-CISPO with causal-tangent saliency, scalar CISPO clipping, and bounded mean-one token weights.
6. CW-CISPO with entropy-only weights.
7. CW-CISPO with causal-tangent saliency multiplied by entropy capacity.
8. CW-CISPO or AD-CISPO-Bound with exact future-attention in-degree saliency and KV-norm proxy saliency.
9. Shuffled-saliency, shuffled-entropy, and uniform-weight controls.
10. Hard selective-CISPO top- p token ablation.
11. Optional adaptive-GRPO ablation, clearly separated from AD-CISPO.

Metrics. Report both task performance and optimizer behavior:

- exact-match accuracy, verifier reward, and pass@k by dataset and difficulty tier;
- reward-per-token and reward-per-GPU-hour;
- upper-clip saturation rate and token update coverage;
- entropy, approximate KL to the reference model when logged, and response length;
- CW-CISPO weight mean, min, max, saturation fraction, and entropy concentration;
- AD-CISPO multiplier mean, min, max, saliency mean, and saliency concentration for bound-only and hybrid variants;
- high-saliency token category frequencies, including comma, punctuation, numeric, mathematical-operator, and answer-token fractions;
- high-weight token category frequencies, separately from high-saliency frequencies when entropy gating is enabled;
- sink-guard removed-token fraction and removed-saliency-mass fraction;
- saliency-extractor overhead in wall-clock time and peak memory, separately for target-policy causal-tangent and frozen-reference baselines.

Decision rule. CW-CISPO should be considered successful only if it improves exact-match or reward efficiency over scalar CISPO at comparable compute, or if it provides a clear stability advantage such as lower collapse rate or smoother reward improvement. A gain that also appears for shuffled saliency, entropy-only weights, or uniform weights should be treated as evidence for generic regularization, uncertainty weighting, or changed effective learning rate, not for the proposed causal-token mechanism. AD-CISPO-Bound should be retained only if it beats both scalar CISPO and CW-CISPO, because bound-only adaptation is the weaker mechanistic lever when policy ratios rarely reach the upper bound.

14 Milestones

1. **Week 1:** verify MiniMax-style CISPO against GRPO on GSM8K with matched model, batch size, rollout count, and reward function.
2. **Week 2:** implement detached mean-one token-credit weights in the CISPO loss; support uniform, shuffled, entropy-only, causal-tangent, and causal-tangent-times-entropy weight sources.
3. **Week 3:** run scalar CISPO, AD-CISPO-Bound, CW-CISPO causal-tangent, CW-CISPO entropy-only, and CW-CISPO hybrid saliency-entropy ablations with matched seeds.
4. **Week 4:** analyze reward curves, clipping dynamics, token-weight distribution, saliency concentration, sink-guard behavior, runtime overhead, and representative trajectory changes.

15 Risks, Ablations, and Fallbacks

Risk: attention saliency is not causal. Attention weights may not explain model behavior [8]. This is why the next-generation plan treats future-attention in-degree as a diagnostic baseline and uses causal-tangent saliency as the primary extractor. The shuffled saliency ablation remains required to verify that any gain is coming from the token-importance signal rather than from weight or multiplier variance.

Risk: attention sinks distort saliency. Initial and special tokens can attract attention for non-semantic reasons [9]. The implementation must mask prompt and sink tokens, use generated-token-only normalization, and report saliency concentration diagnostics. For causal-tangent saliency, sink guards should be kept explicit rather than hidden inside the saliency formula so that the removed token and removed-mass fractions can be audited.

Risk: causal-tangent overhead dominates. Causal-tangent saliency adds an extra objective-specific backward pass. The fallback is to compute it every N rollout batches, use only the final two decoder layers, reduce the saliency minibatch size, or fall back to KV-norm or future-attention in-degree for cheaper diagnostic runs.

Risk: saliency weighting changes the effective learning rate. Even mean-one token weights can alter gradient variance and interact with minibatch composition. The fallback is to keep the response-level mean fixed, track per-minibatch gradient norms, and compare against uniform random weights with the same marginal distribution. If the random-weight control matches CW-CISPO, the gain is not evidence for causal-token credit.

Risk: entropy explains the gain. If entropy-only CW-CISPO matches causal-tangent CW-CISPO, the saliency extractor is unnecessary for the next version. In that case, the project should pivot to Entropy-Capacity CISPO as the simpler CISPO-native extension, while keeping causal saliency as a diagnostic.

Risk: budget concentration increases variance. Large multipliers may overemphasize a small set of tokens. The fallback is a bounded weight or multiplier projection with post-clamp renormalization, plus reporting clip-min, clip-max, weight-min, weight-max, and saturation metrics.

16 Expected Impact

If successful, this CISPO-extension program would provide evidence that reasoning-RL trust regions and token-gradient mass should be token-budgeted rather than uniformly scalar. The contribution would be useful even if the exact causal-tangent extractor is later replaced: the broader idea is to keep CISPO’s clipped-IS stability while adding a falsifiable token-credit layer derived from the current policy’s own rollout computation. A negative result would still be informative because it would test whether CISPO’s scalar gradient allocation is already sufficient, whether entropy alone explains token credit in RLVR, and whether mechanistic saliency signals are too weak or too noisy for optimizer control.

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